

BraTS MRI Reconstruction

Multi-contrast MRI Reconstruction from Undersampled Data

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Project Motivation

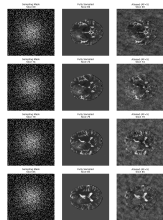
- Accelerated MRI acquisition is clinically desirable, but k-space undersampling introduces aliasing artifacts.
- Objective: reconstruct high-fidelity T2 MRI slices from $AF=5$ undersampled data.
- Study structure: simulation, baseline reconstruction, and multi-modal unrolled reconstruction.

Dataset and Experimental Setup

- Dataset: BraTS, using co-registered T1 and T2 modalities.
- Split strategy: patient-level partition to prevent train/test leakage.
- Final formal run: 16 central slices per patient across all available patients.
- Runtime optimization: slice caching, pinned memory, TF32, and non-blocking GPU transfer.

Methods Overview

Task 1: Undersampling Simulation



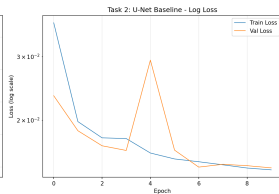
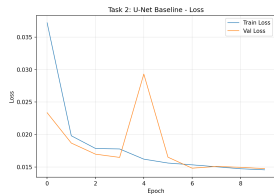
Task 2: Baseline U-Net

- Input: undersampled T2 slice.
- Model: U-Net baseline.
- Loss: MSE.

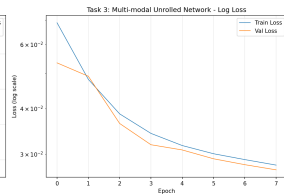
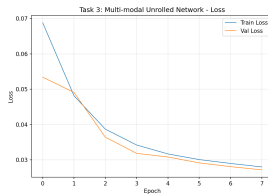
Task 3: Multi-modal Unrolled Model

- Input: undersampled T2 + fully sampled T1.
- Model: 2-cascade unrolled U-Net with data consistency.
- Loss: hybrid L1/L2.

Training Behaviour



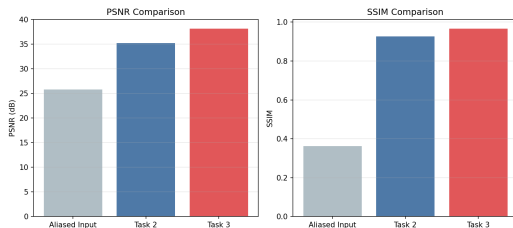
Task 2 final performance: PSNR 35.18, SSIM
0.9271



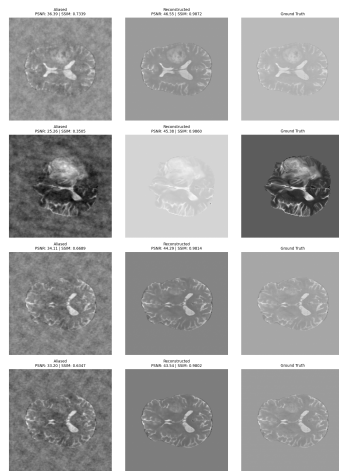
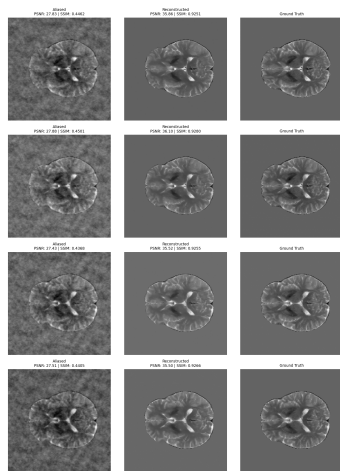
Task 3 final performance: PSNR 38.12, SSIM
0.9660

Quantitative Comparison

- Task 2 achieves PSNR 35.18 and SSIM 0.9271.
- Task 3 achieves PSNR 38.12 and SSIM 0.9660.
- Task 3 exceeds Task 2 by 2.93 dB PSNR.
- Task 3 exceeds Task 2 by 0.0389 SSIM.

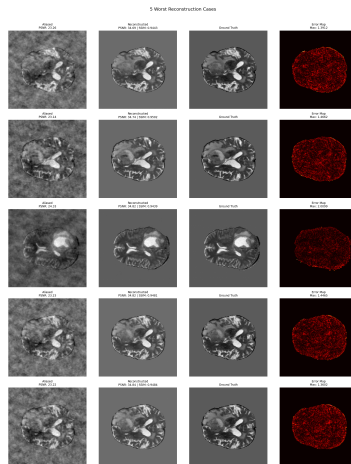


Representative Reconstructions



Task 3 Error Analysis

- The lowest-performing cases are concentrated in structurally complex slices.
- Even these cases remain substantially better than the aliased input.
- Likely improvements: stronger edge-aware losses, deeper cascades, and hard-case-focused training.



Conclusions

- The baseline model removes most of the aliasing artifacts effectively.
- The multi-modal unrolled model provides the best quantitative fidelity and visual quality.
- T1 structural guidance and data consistency together yield measurable gains over the baseline.
- Future work includes stronger edge-aware objectives, deeper cascades, and targeted hard-case training.

AI Usage Declaration

- AI tools were used only for debugging support, experiment automation, and draft editing.
- All modelling choices, metric verification, result interpretation, and submitted materials were manually reviewed and confirmed.
- This declaration is included to satisfy the assignment requirement.